

Homework 8

Due: Thursday 5/15 at 11:59pm

Complete the following problems. Submit the following through Google Classroom. Be sure to hit the Submit button.

1. (21 points) Use recursion to implement each of the following methods in `RecursiveList`, building on the code from Lab 7:
 - (a) A method `numBeforeK` that takes an integer `k` and a list value. It returns the number of times this value occurs in the first `k` list locations. If `k` is 0 or less, the method should return 0. If `k` is greater than the number of values in the list, it should return the total number of occurrences. Note that `k` is the number of values to examine, not the index up to which to examine the list.
 - (b) A method `addAfter`, which takes two arguments and adds the first argument to the list after the first occurrence of the second argument. If the second argument doesn't appear in the list, then `addAfter` does nothing.
 - (c) A method `remove` that takes an index and removes the value at that index. If the index is out of bounds, the method instead throws an `IndexOutOfBoundsException`.

Note that you should write test cases to ensure these methods work, but you will not be graded on the test cases themselves.

2. (1 point) At the top of the `RecursiveList.java` file, include a comment with a statement about your collaboration (if any).